NATIONAL INSTITUTE OF TECHNOLGY PATNA
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
MID-SEMESTER EXAMINATION – ODD SEMESTER 2025

III Semester CS34104 – Data Structures Max.Marks:30

Answer all questions

Q. No	Question	Marks	CO & BL

1.	Find the worst case time complexity of the following:
	a)
	$for\ (i = 1;\ i^3 \le n; i++)$
	{
	$for(j = n; j \ge 1;\ j = j/3)$
	{
	$for(k = 1; k^{27} < n^5; k++)$
	{
	$a = a + b;$
	}
	}
	}
		6M	CO-1 & Evaluate

	b)
	$main()$
	{
	$i = 2;$
	$While(i < n)$
	{
	$i *= i;$
	$i *= i *= i;$
	$i += 2;$
	$i *= 2;$
	}
	}		

	c)
	$main()$
	{
	$i = n;$
	$While(i > 1)$
	{
	$i = i^{1/2};$
	}
	}		

2.	Design an efficient algorithm to check whether a given singly link list has a cycle? Find the time complexity of your algorithm.	4M	CO-1 & Design
3.	a) Design quicksort algorithm to sort a list of items in non-decreasing order. The algorithm should also calculate and returns the numbers of comparisons and movements. b) Apply the algorithm designed in the above question to sort the following list $\{E, X, A, M, P, L, E, C, I, V, E, N\}$ in alphabetical order show all the steps.	5M	CO-4 Design & Apply

4.	Translate the following infix expression into a postfix expression using stack. Must show the contents of the stack and partial output after reading every character from the infix expression. Infix Expression: : $x = (a - b + 1) \times c - d \wedge e \wedge f + g$ Evaluate the postfix expression obtained from the above question by using stack. Consider the current value of the variable as: $x = 134,\ a = 7,\ b = 6,\ c = 4,\ d = 2,\ e = 3,\ f = 2,\ g = 5$	6M	CO-2 & Apply
5.	The *input restricted double ended queue* is a queue in which the insertion operation is allowed in one end (typically the rear) while deletion can be perform from both ends (rear and front). Design the procedures for insertion and deletion operations for an input restricted double ended queue, must include the condition for overflow and underflow.	4M	CO-2 & Design

6.	Find the tightest asymptotic time complexity, of the following function:
	`void MSE(int fun)`
	`{`
	`if (fun>0)`
	`{`
	`MSE(fun-1);`
	`printf("%d", fun);`
	`MSE(fun-1);`
	`}`
	`}`
	Find the number of invocations takes place and output of the above program if the initial function call is: `MSE(5)`.
		5M	CO-1 & Analyse & Evaluate

********** ALL THE BEST **********